

Evaluating, Synthesizing, and Enhancing for Customer Support Conversation

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Abstract

Effective customer support requires not only accurate problem-solving but also structured and empathetic communication aligned with professional standards. However, existing dialogue datasets often lack strategic guidance, and real-world service data is difficult to access and annotate. To address this, we introduce the task of Customer Support Conversation (CSC), aimed at training customer service supporters to respond using well-defined support strategies. We propose a structured CSC framework grounded in COPC guidelines, defining five conversational stages and twelve strategies to guide high-quality interactions. Based on this, we construct *CSCConv*, an evaluation dataset of 1,855 real-world customer-agent conversations rewritten using LLMs to reflect deliberate strategy use, and annotated accordingly. Additionally, we develop a role-playing approach that simulates strategy-rich conversations using LLM-powered roles aligned with the CSC framework, resulting in the training dataset *RoleCS*. Experiments show that fine-tuning strong LLMs on *RoleCS* significantly improves their ability to generate high-quality, strategy-aligned responses on *CSCConv*. Human evaluations further confirm gains in problem resolution. All code and data will be publicly available at <https://github.com/aliyun/qwen-dianjin>.

1 Introduction

Customer support aims to help users resolve product- or service-related issues through effective and context-aware communication. While large-scale dialogue systems have received growing attention in recent years (Budzianowski et al. 2018; Rashkin et al. 2019; Peskov et al. 2019; Liu et al. 2021), customer support conversations remain underexplored in the NLP community,¹ largely due to the scarcity of publicly available benchmarks and the sensitive, domain-specific nature of support interactions. Meanwhile, large language models (LLMs) like GPT-3 (Brown et al. 2020) have shown impressive capabilities in open-domain dialogue generation (Zhang et al. 2020b; Bae et al. 2022; Thoppilan et al. 2022), but their ability to generate realistic and effective customer support conversations remains underexamined.

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¹This paper uses *dialogue* and *conversation* interchangeably.

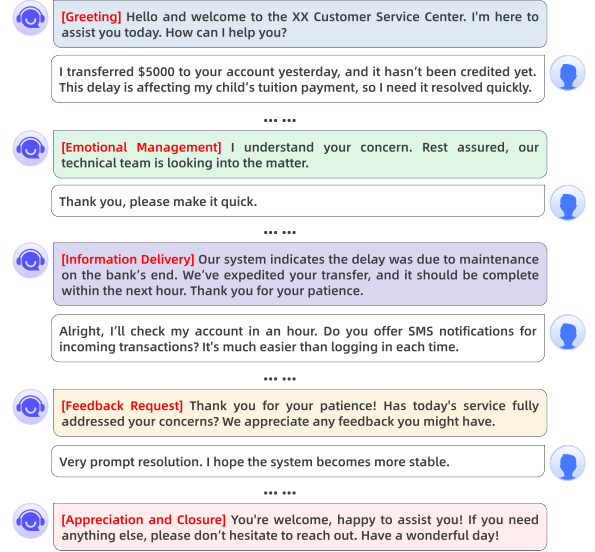


Figure 1: An example dialogue between a service supporter (left) and a customer (right), showing **support strategies** (noted in parentheses) used by the supporter. The conversation is organized into five stages of the proposed CSC framework, shown in colored boxes.

Customer support guidelines from COPC, the internationally recognized standard for customer experience management,² emphasize that high-quality support often depends on structured communication strategies, similar to those used in emotional support settings (Liu et al. 2021). As shown in Figure 1,³ the service supporter begins with the *Greeting* strategy to initiate the conversation and explore the customer's issue. Upon recognizing the customer's negative emotion, the supporter adopts the *Emotional Management* strategy, expressing empathy and understanding to alleviate the customer's distress. After understanding the problem, the supporter employs the *Information Delivery* strategy to provide clear and actionable guidance, followed by the *Feedback Request* strategy to check for further concerns. Finally, the conversation ends with the *Appreciation and Closure*

²<https://www.copc.com/copc-standards/>

³The example is translated into English for better readability.

strategy, ensuring a respectful and reassuring followed by *Feedback Request* to check for further concerns.

Despite the importance of effective customer support, research on real-time customer service conversations remains limited, mainly due to a lack of task-specific design and high-quality annotated data. Many studies (Xu et al. 2017; Oraby et al. 2017; Cui et al. 2017; Mesquita, Martins, and Almeida 2022) rely on asynchronous, exchanges (e.g., Twitter), where interactions span minutes to days. These settings differ significantly from the immediate, uninterrupted nature of real-time support. Moreover, effective support requires not only resolving issues but also showing empathy and emotional support. Yet, most task-oriented dialogue datasets (Wen et al. 2017; Budzianowski et al. 2018; Peskov et al. 2019; Gung et al. 2023) lack the intentional use of supportive strategies like emotional management or empathetic closure, which are vital for high-quality customer service.

To address this gap, we introduce the task of customer support conversation (CSC), to facilitate the training of customer service supporters. The goal is to help them respond with appropriate strategies that combine accurate solutions with empathetic communication. Based on COPC standards and inspired by the emotional support framework (Liu et al. 2021), we develop a CSC-specific framework with five dialogue stages and twelve support strategies. Using it, we build CSConv, a high-quality dataset adapted from real service interactions and refined for structured strategy use. To address the lack of high-quality training data, we also develop a role-playing approach that creates RoleCS, a synthetic dataset of strategy-rich conversations by assigning LLMs distinct roles aligned with the CSC framework. Fine-tuning on RoleCS significantly boosts LLMs’ ability to generate strategy-aligned and effective responses on CSConv.

2 Related Work

2.1 Task-Oriented Conversation Datasets.

The goal of CSConv, which is to generate conversations based on real spoken customer service interactions, differs significantly from previous task-oriented dialogue datasets. Many earlier synthetic datasets have been created using the Wizard of Oz (WOZ) framework (Kelley 1984), where one person acts as the system and another as the user. Wen et al. (2017) introduce a crowdsourced version of the WOZ framework to collect domain-specific dialogue data more efficiently. Other task-oriented dialogue datasets include Frames (El Asri et al. 2017), MultiWOZ (Budzianowski et al. 2018), MultiDoGO (Peskov et al. 2019), EMPATHET-ICDIALOGUES (Rashkin et al. 2019), Taskmaster-1 (Byrne et al. 2019), ESConv (Liu et al. 2021), and NATCS (Gung et al. 2023), each of which explores different domains and annotation schemes for developing conversational agents.

Among them, ESConv is closely related to our work, as both it and CSConv aim to enhance service quality using structured support strategies. However, their construction methods differ significantly: ESConv uses a crowdsourced WOZ setup, while CSConv rewrites real-world service dialogues using high-performing LLMs.

2.2 Role-Playing in Conversation Generation

Recent advances in LLMs have enabled the use of role-playing agents for conversation generation. For example, Bae et al. (2022) generate dialogues aligned with specific roles, while Yang et al. (2024b) simulate characters and settings to produce coherent interactions. Wu et al. (2024) create interactive narrative dramas, and Ye et al. (2025) focus on emotional support conversations. Building on the Ye et al. (2025), we design role-playing agents for customer support with clearer role definitions and responsibilities to more closely reflect real-world service interactions.

3 CSC: Customer Support Conversation

In customer support conversation (CSC), the service supporter aims to accurately identify customer issues while also addressing emotional needs. This dual focus enables effective solutions that enhance customer satisfaction and improve problem-resolution efficiency. The supporter is considered effective when both the issue and emotional concerns are properly addressed. To build an evaluation dataset for CSC, we first introduce the CSC framework in Section 3.1, followed by the data construction process in Section 3.2 and dataset statistics in Section 3.3. Finally, we define the CSC task in Section 3.4.

3.1 CSC Framework

The CSC framework organizes the customer support process into five stages, each with recommended strategies designed to enhance the quality and effectiveness of interaction.

Stages. Building on the three core stages of supportive communication from Hill (2019) and the COPC practical guidelines, we work with domain experts to refine and extend this structure for customer support. The resulting CSC framework defines five stages: *Connecting* (greeting and rapport building), *Identifying* (understanding the customer’s issue and emotional state), *Exploring* (discussing and evaluating potential solutions), *Resolving* (delivering and confirming resolution), and *Maintaining* (closing the interaction while preserving the customer relationship). Figure 2 illustrates the flow across these stages.

Importantly, these stages are not rigid steps but modular components that can appear in various orders or combinations depending on the nature of the conversation. For example, even when no solution is reached, a supporter may still provide empathy (*Exploring*), acknowledge service limitations (*Resolving*), and close the conversation professionally (*Maintaining*). This flexible structure enables consistent analysis and strategy modeling across both successful and challenging customer support scenarios.

Strategies. In parallel, we work with domain experts to define twelve actionable support strategies aligned with the five stages: *Greeting (GT)*, *Identity Verification (IV)*, *Emotional Management (EM)*, *Restatement or Paraphrasing (RP)*, *Problem Refinement (PR)*, *Providing Suggestions (PS)*, *Information Delivery (ID)*, *Resolution Implementation (RI)*, *Feedback Request (FR)*, *Appreciation and Closure (AC)*, *Relationship Continuation (RC)*, and *Others*.

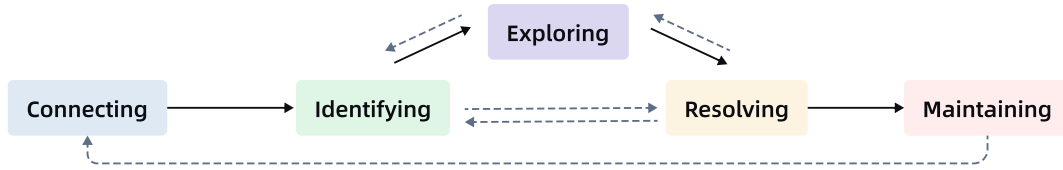


Figure 2: Overview of the CSC framework’s five stages, each paired with recommended support strategies (see Table 9 in Appendix A). The typical flow is: ① Connecting → ② Identifying → ③ Exploring → ④ Resolving → ⑤ Maintaining (black arrows), but it can be adjusted based on the specifics of each conversation (dashed arrows).

These strategies provide practical guidance for managing both task-related issues and emotional support throughout the customer interaction. For the details of stages and strategies, please refer to Appendix A.

3.2 Dataset Construction

We construct the *CSCConv* dataset using Chinese customer service dialogues collected from our pre-sales and after-sales customer service centers. Prior to our access, all conversations were professionally transcribed, manually corrected, and fully de-identified to ensure privacy protection and integrity. While these raw conversations authentically reflect real-world customer interactions, they often lack consistent structure, making it difficult to systematically annotate support strategies according to the COPC-informed CSC framework. To address this, we employ an LLM to rewrite the conversations while preserving the original semantics and user intent. This controlled rewriting aligns conversations with the defined dialogue stages and strategies, enabling more consistent and interpretable annotations without sacrificing the complexity of real user queries.

Importantly, the goal of constructing *CSCConv* is not to evaluate real-time chatbot performance, but to facilitate the training of customer service supporters by helping them learn to respond using appropriate strategies guided by the COPC framework. In total, we collect 690K conversations spanning eight in-domain topics. To ensure quality, we use a four-stage pipeline to guide data selection and refinement.

1. **Pre-filtering:** We first apply rule-based filtering to remove low-quality conversations. These rules exclude conversations that are too short or too long, contain overly lengthy utterances, show an imbalance between customer and agent turns, or have a high proportion of ineffective customer responses. Additionally, we use an LLM to exclude conversations with offensive or unprofessional content.
2. **Sampling and Rewriting:** Up to 500 filtered conversations per topic are sampled and rewritten by an LLM to align with the CSC framework. During this process, the LLM analyzes the original scenario and generates a new conversation that preserves the core issue while improving clarity, structure, and emotional engagement. For each agent turn, the LLM selects an appropriate support strategy based on conversation context, occasionally using *Others* to maintain conversational naturalness. Customer responses are also refined for coherent interaction.
3. **Post-filtering:** After rewriting, a second round of rule-based and LLM-based checks verifies structure (e.g.,

strategy coverage, speaker alternation, and length) and filters out conversations lacking coherence, empathy, or strategic alignment.

4. **Manually annotation:** Finally, experts certified in COPC review the remaining conversations, evaluating the support agent’s responses for realism, empathy, and adherence to the CSC framework. This results in a curated evaluation set of 1,855 high-quality conversations.

Appendix B provides details on filtering rules, prompt templates, and annotation guidelines. For rewriting, we use *DeepSeek-R1* (Guo et al. 2025) instead of *GPT-4o* (OpenAI 2024), as the latter tends to produce shorter, less emotionally rich dialogues. See Appendix B.5 for a comparison.

3.3 Dataset Analysis

Overview Statistics. Table 1 compares statistics of *CSCConv* before and after rewriting, covering 1,855 conversations. Rewritten dialogues are longer, averaging 27.27 versus 19.06 utterances, with supporter responses increasing from 41.16 to 48.72 words and customer responses decreasing from 21.60 to 17.17. Notably, strategy use (excluding *Others*) rises from 55.28% to 97.82%,⁴ indicating more deliberate and guided support. These shifts reflect the goal of rewriting, which is to facilitate the training of customer service supporters in applying appropriate strategies aligned with the COPC framework.

Topic Distribution. Table 2 presents the distribution of conversations across eight topic. Each topic, excluding *Others*, accounts for roughly 11% to 16% of the dataset.

Strategy Distribution. As shown in Figure 3, the most commonly used strategies are *Information Delivery* (14.9%), *Emotional Management* (11.9%), and *Provide Suggestions* (10.0%). This highlights the dual importance of delivering clear information and managing customer emotions in effective support conversations.

Additional statistics on strategy transitions and unique strategy distributions are provided in Appendix C.

3.4 CSC Task Definition

We denote a customer support conversation as $D = \{(P_i, T_i, U_i)\}_{i=1}^N$, consisting of N turns exchanged between a supporter S and a customer C . For each turn, $P_i \in \{S, C\}$

⁴We use *Qwen2.5-72B-Instruct* to assign a support strategy to each supporter response in the dialogues prior to rewriting.

		Number	
		Original	Rewritten
Total	Conversations	1,855	1,855
	Utterances	35,350	50,587
	Avg. Utterance Number	19.06	27.27
	Avg. Utterance Length	31.48	33.27
Supporter	Utterances	17,862	25,810
	Avg. Utterance Number	9.63	13.91
	Avg. Utterance Length	41.16	48.72
	Strategy Use (w/o <i>Others</i>)	55.28%	97.82%
Customer	Utterances	17,488	24,777
	Avg. Utterance Number	9.43	13.36
	Avg. Utterance Length	21.60	17.17

Table 1: Statistics of CSConv before and after rewriting.

Topic	Num	Proportion
Account and Transaction Management	265	14.3%
Product Consultation	242	13.0%
Technical Support and Online Services	295	15.9%
Complaints and Dispute Resolution	263	14.2%
Marketing and Promotion Activities	211	11.4%
Risk Management and Security	266	14.3%
Financial Consulting and Planning	263	14.2%
Others	50	2.7%
Overall	1,855	100.0%

Table 2: Distribution of topics in CSConv.

denotes the speaker, U_i is the utterance text, and T_i represents the response strategy used. Strategies are selected from a predefined set G , and are assigned only to the supporter’s turns. That is, if $P_i = C$, then $T_i = \text{NULL}$.

Following the task of emotional support conversation (Liu et al. 2021; Ye et al. 2025), we define the CSC task as generating the supporter’s response. Specifically, at turn k , where $P_k = S$, the model receives the conversation history $X_k = \{(P_i, T_i, U_i)\}_{i=1}^{k-1}$ as input. The CSC task is then divided into two subtasks:⁵

1. **Strategy Prediction:** Predict the appropriate support strategy $T_k \in G$ based on the conversation history X_k .
2. **Response Generation:** Generate a response U_k conditioned on both the predicted strategy T_k and the conversation history X_k , ensuring that the output is consistent with both the customer’s needs and the strategic intent.

4 Synthetic Conversation Generation with Role-Playing Agents

Fine-tuning LLMs has shown effective for improving performance in task-specific dialogue applications. However, building high-quality, multi-turn customer support datasets remains challenging due to the need for domain expertise, contextual coherence, and dialogue diversity. This limits both the scalability and scenario coverage. While recent studies use LLMs for dataset augmentation via rewriting or imitation, they often yield limited variation in dialogue

⁵We further investigate how strategy prediction contributes to improving response generation in Section 6.3.

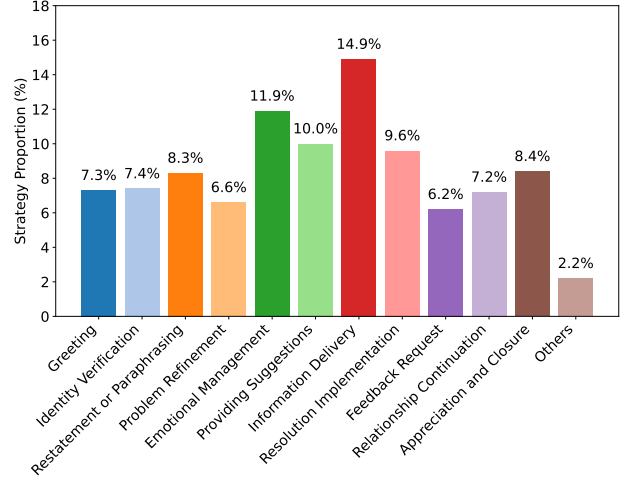


Figure 3: Strategy proportion of CSConv.

flow and support strategy. To address this, inspired by Ye et al. (2025), we adopt a multi-role role-playing framework to generate synthetic customer support dialogues that are diverse, coherent, and representative of real-world scenarios.

4.1 Role Playing Conversation Generation

As shown in Figure 4, our role-playing framework involves five roles: *Planner*, *Supporter Assistant*, *Supporter*, *Customer Assistant*, and *Customer*. Each role serves a distinct purpose. The *Planner* defines the dialogue scenario and sets the *Customer*’s communication goal. The main interaction occurs between the *Supporter* and the *Customer*, simulating a real customer service exchange. Meanwhile, their respective *Assistants*, the *Supporter Assistant* and the *Customer Assistant*, offer strategic guidance to help them fulfill their roles more effectively. We use deepseek-r1 (Guo et al. 2025) to simulate all roles, with detailed prompts available in Appendix E. We describe each role below.

Planner Before the conversation begins, the Planner selects a topic e from a predefined list E of customer topics in Table 2 (excluding *Others*). It then samples a customer profile o from a character pool O that contains diverse customer personas. Using the selected topic e and customer profile o , the Planner prompts an LLM $\mathcal{M}$ to generate a detailed service scenario e' along with a corresponding communication goal g , denoted as $(g, e') = \mathcal{M}(o, e)$. This setup ensures that each dialogue is grounded in a realistic context.

Supporter Assistant The Supporter Assistant recommends an appropriate support strategy t from the predefined strategy set G , based on the current dialogue history h_s and the scenario e' . The recommendation is generated by querying the LLM $\mathcal{M}$, i.e., $t = \mathcal{M}(h_s, G, e')$. This strategic guidance helps the Supporter stay aligned with the customer’s needs and maintain coherent throughout the conversation.

Supporter Guided by the support strategy t suggested by the Supporter Assistant, the Supporter generates a contextu-

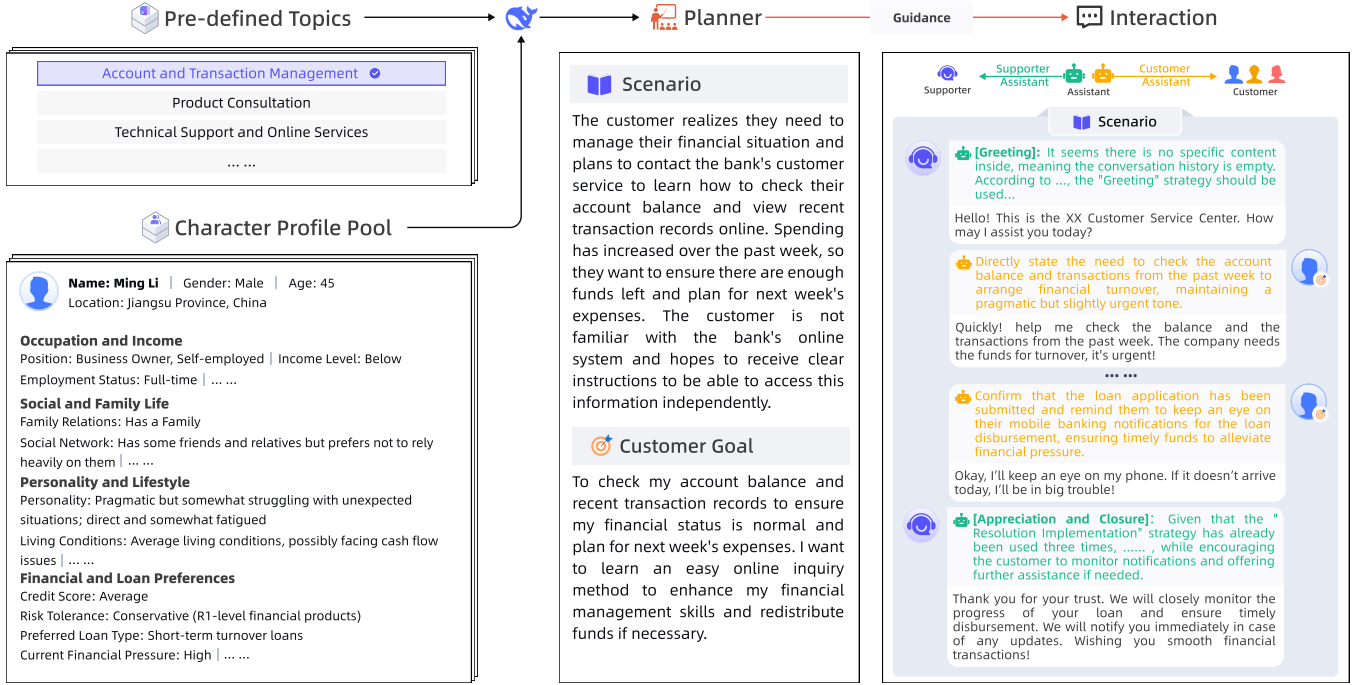


Figure 4: Illustration of synthetic conversation generation using role-playing agents.

ally appropriate response r_s . This response is produced by the LLM $\mathcal{M}$, conditioned on the dialogue history h_s , the strategy t and the scenario e' , i.e., $r_s = \mathcal{M}(h_s, t, e')$.

Customer Assistant The Customer Assistant guides the conversation by generating the next direction d , ensuring alignment with the customer’s communication goal g . To do so, it prompts the LLM $\mathcal{M}$ using the current dialogue history h_c , the customer’s goal g , and the scenario e' . This helps the customer maintain coherent and goal-oriented behavior throughout the conversation. Formally, the direction is generated as $d = \mathcal{M}(h_c, g, e')$.

Customer The Customer generates a response r_c based on the current dialogue history h_c , the intended conversational direction d , the character profile o , and the scenario e' . This ensures that the response is both contextually appropriate and consistent with the customer’s persona. Formally, the generation process is denoted as: $r_c = \mathcal{M}(h_c, d, o, e')$.

Diversity in conversations relies heavily on the richness of customer personas (Wang et al. 2025). To support this, we construct a rich character profile pool O that captures a wide spectrum of customer backgrounds, behaviors, and communication styles, thereby enhancing the realism and variability of the simulated interactions.

Character Profile Pool. We design a comprehensive character profile template specifically for customer personas, incorporating attributes such as demographics, financial status, and communication preference. To populate this template, we use Qwen2.5-72B-Instruct to automatically extract and complete profile information from 15,980 real-world customer service dialogues. To reduce redundancy

and ensure profile diversity, each structured profile is converted into a free-text description, and pairwise cosine similarity is computed using Qwen’s `text-embedding-v2`. Profiles exceeding a similarity threshold of 0.85 are considered redundant and removed. Following this filtering process, we retain 1,948 distinct customer profiles in our final pool. Additional details, including the profile template and construction prompts, are provided in Appendix D.

4.2 Synthetic Dataset

The Planner generates one conversation per unique pair of customer topic and profile. With $(|E| - 1)$ topics (excluding *Others*) and $|O|$ profiles, this yields 13,636 conversations. After applying quality filters detailed in Appendix F, we retain 11,232 high-quality conversations, forming the final training (or fine-tuning) dataset, `RoleCS`. To assess whether customer utterances reflect assigned profiles, we randomly sample 500 dialogues and measure word overlap between customer utterances and their corresponding aligned profiles, using random profiles for comparison. As shown in Figure 5, word overlap with aligned profiles rises rapidly at first as profile information is elicited, then increases more gradually once most information has been mentioned. Throughout, word overlap with aligned profiles remains consistently higher than with random profiles, indicating that customer utterances naturally reflect profile details. See Appendix G for more detailed analysis of `RoleCS`.

5 Experimentation

Test-Time Prompting. For both subtasks outlined in Section 3.4, we adopt a unified single-prompt method. Given the

Model	Size	B-2	B-4	R-L	BS	BR	ACC	B-2	B-4	R-L	BS	BR	ACC
<i>Evaluation with reference context</i>							<i>Evaluation with generated context</i>						
GPT-4o	-	8.13	2.97	4.12	64.26	51.27	42.58	6.41	2.07	2.22	62.95	51.48	36.29
DeepSeek-R1	671B	11.67	5.09	8.44	66.57	52.09	39.78	8.41	3.11	6.27	64.80	52.01	35.23
DeepSeek-V3	671B	11.57	4.95	7.04	66.09	53.10	41.99	8.43	3.33	4.14	64.13	<u>53.09</u>	<u>36.54</u>
LLaMA3.1-Instruct	8B	4.28	1.44	3.62	58.68	38.84	17.16	2.62	0.89	1.31	55.06	<u>35.97</u>	<u>13.75</u>
+ RoleCS	8B	11.06	4.77	6.93	66.21	52.13	42.15	8.80	3.72	4.52	64.57	49.85	35.73
LLaMA3.1-Instruct	70B	6.85	2.38	4.10	63.14	47.53	38.78	5.57	1.76	2.59	62.24	<u>46.83</u>	<u>30.36</u>
+ RoleCS	70B	11.73	5.11	7.64	66.62	53.69	42.79	9.62	4.00	<u>4.89</u>	65.15	50.91	39.44
Qwen2.5-Instruct	7B	6.73	2.29	3.80	62.75	48.21	18.78	5.11	1.53	2.43	61.62	47.43	19.39
+ RoleCS	7B	11.23	4.80	7.39	66.35	52.14	42.96	8.61	3.51	4.13	64.42	48.80	30.52
Qwen2.5-Instruct	72B	8.61	3.23	5.41	64.63	52.49	37.22	6.28	2.00	3.59	63.14	<u>50.55</u>	<u>30.93</u>
+ RoleCS	72B	12.15	5.32	<u>7.97</u>	66.85	54.49	43.29	<u>9.38</u>	<u>3.90</u>	4.35	<u>64.89</u>	53.65	36.02

Table 3: Performance comparison on CSConv. The best and second best results are in **bold** and underlined, respectively.

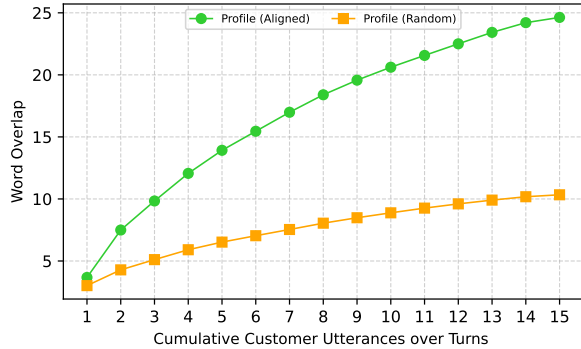


Figure 5: Word overlap between cumulative customer utterances and profile (Aligned vs. Random).

conversation history, the LLM, whether fine-tuned or not, is prompted once to first identify the appropriate support strategy and then generate the corresponding response. The full prompt is provided in the Appendix I.

5.1 Experimental Settings

Models. We evaluate several widely used LLMs on the CSC task, including GPT-4o (OpenAI 2024), DeepSeek-R1 (Guo et al. 2025), DeepSeek-V3 (Liu et al. 2024), Qwen-2.5-7B/72B-Instruct (Yang et al. 2024a), and LLaMA-3.1-8B/70B-Instruct (Grattafiori et al. 2024). To evaluate the impact of the proposed RoleCS, we fine-tune the Qwen and LLaMA models using 137,406 fine-tuning instances extracted from RoleCS, formatted consistently with our single-prompt in test-time. Fine-tuning and inference configurations are provided in Appendix J.

Metrics. Following prior work (Liu et al. 2021; Ye et al. 2025), we adopt a diverse set of metrics to assess the response quality. These include BLEU-n (**B-n**) (Papineni et al. 2002) and ROUGE-L (**R-L**) (Lin 2004) for measuring lexical overlap, BERTScore (**BS**) (Zhang et al. 2020a) and BLEURT (**BR**) (Sellam, Das, and Parikh 2020) for semantic similarity. In addition, to evaluate alignment with the intended support strategy, which is a core aspect of the CSC

task, we report strategy prediction accuracy (**ACC**).

5.2 Experimental Results

Table 3 presents the main results on CSConv under two evaluation settings: (1) *evaluation with reference context*, which all LLMs are evaluated using the same gold history for fair comparison, and (2) *evaluation with generated context*, which assesses performance when relying on model-generated histories, thus reflecting the ability to maintain coherence and relevance without ground truth context. From the results, we have the following observations:

- Larger models tend to perform better among non-fine-tuned LLMs. Additionally, Chinese-centric models like Qwen and DeepSeek outperform more general models such as LLaMA and GPT, indicating that alignment with language and cultural context benefits CSC performance.
- RoleCS proves highly effective, as fine-tuning on it significantly improve performance across all metrics. In particular, Qwen2.5-Instruct-72B, after fine-tuning, matches or surpasses DeepSeek-R1, a strong baseline for Chinese-language tasks.
- Evaluation with generated context shows similar performance trends as with reference context, but with lower absolute scores. This drop reflects the difficulty of maintaining consistency and quality in multi-turn conversations when relying on model-generated dialogue history, echoing findings in prior work such as Ye et al. (2025).

6 Discussion

We conduct further analysis under the reference context setting to examine the effectiveness of our role-playing approach in synthetic conversation generation.

6.1 Impact of Role-Playing on Data Quality

To evaluate our role-playing framework, we compare RoleCS with two baselines:

- Baseline 1: Conversations are generated via in-context learning without any role-playing.
- Baseline 2: Role-playing is applied, but without the Supporter Assistant agent.

SFT Data	B-2	B-4	R-L	BS	BR	ACC
Baseline 1	8.51	3.09	5.35	64.35	46.70	33.09
Baseline 2	9.40	3.26	6.98	65.22	50.18	36.17
RoleCS	11.23	4.80	7.39	66.35	52.14	42.96

Table 4: Performance comparison of fine-tuning on different datasets.

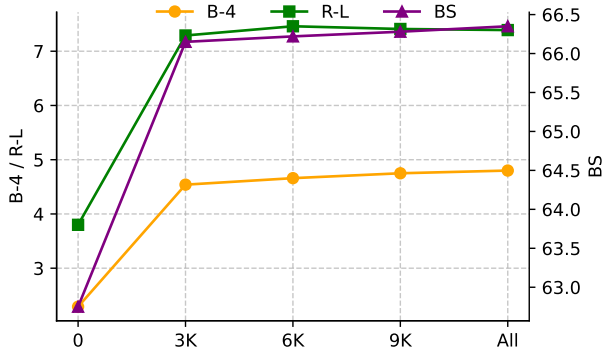


Figure 6: Performance comparison under different synthetic conversation dataset sizes.

All three datasets are generated using Deepseek-R1 with equal conversation counts. For fairness, we fine-tune Qwen2.5-7b-Instruct on each. As shown in Table 4, role-playing improves performance over in-context learning (Baseline 2 vs. Baseline 1), and adding the Supporter Assistant further boosts results (RoleCS vs. Baseline 2). These results highlight the value of strategy-rich training data and the importance of high-quality dataset construction.

6.2 Effect of Synthetic Dataset Size

To understand how dataset size influences performance, we split the RoleCS dataset into subsets of {0, 3K, 6K, 9K, All} dialogues. We fine-tune Qwen2.5-7B-Instruct on each subset individually. As shown in Figure 6, the most significant gains occur with the initial 3K examples. Beyond this point, performance improvements become marginal, particularly for ROUGE-L. These results suggest that even a modest amount of high-quality synthetic data can bring strong gains, with limited benefits from adding more.

6.3 Impact of Support Strategy Guidance

We evaluate the effect of incorporating support strategies using three variants of Qwen2.5-72B-Instruct without fine-tuning: (1) Vanilla, where the model generates a response without any support strategy guidance; (2) Predict, our default setup, where the model first predicts the strategy and then generates a response; and (3) Oracle, where the ground-truth strategy is provided in the prompt.

As shown in Table 5, the Predict variant slightly outperforms the Vanilla, indicating that even simple strategy prediction improves response quality. The Oracle variant performs best, highlighting that more accurate strategy prediction can further enhance the performance of CSC task.

Strategy	B-4	BS	ACC
Vanilla	3.11	64.43	-
Predict	3.23	64.63	37.22
Oracle	3.80	66.34	100.00

Table 5: Performance comparison across different support strategy variants.

Model	Size	GPT-4o	Qwen-Plus	Human
DeepSeek-R1	671B	90.89	89.70	3.55
DeepSeek-V3	671B	90.54	89.54	3.36
LLaMA3.1-Instruct Δ	8B	90.34	88.68	2.93
LLaMA3.1-Instruct Δ	70B	91.02	89.76	3.58
Qwen2.5-Instruct Δ	7B	90.46	88.76	3.10
Qwen2.5-Instruct Δ	72B	91.04	89.98	3.79

Table 6: Model performance evaluated by GPT-4o, Qwen-Plus, and human judges. Δ denotes fine-tuned models.

Agreement	Kappa
GPT-4o-Judge & Human Annotators	0.658
Human Annotators	0.628

Table 7: Fleiss’ Kappa scores.

6.4 Evaluation with LLMs and Human as Judges

To mitigate potential bias (e.g., GPT-4o favoring its own outputs), we use both GPT-4o and Qwen-Plus⁶ to evaluate response quality across six dimensions: accuracy, helpfulness, understanding, coherence, informativeness, and empathy, each scored on a 0–100 scale. We report the average overall scores from both GPT-4o-Judge and Qwen-Plus-Judge, using the evaluation prompt in Appendix K. As shown in Table 6, both judges reveal consistent performance patterns. Notably, fine-tuned Qwen2.5-Instruct-72B and LLaMA3.1-Instruct-70B outperform other models, even exceeding DeepSeek-R1 in overall quality.

For human evaluation, we randomly select 100 conversations and have them independently rated by three professional annotators on a 1-5 Likert scale (Joshi et al. 2015) across the same six dimensions. Table 6 shows the average overall scores, which align with trends from the LLM-based evaluations. Table 7 reports Fleiss’ Kappa (Fleiss 1971) scores among annotators and between GPT-4o-Judge and the annotators. The results show strong inter-rater agreement and confirm that GPT-4o-Judge aligns well with human judgment.

7 Conclusion

This paper addresses the challenges of customer support conversations (CSCs) in the NLP field by introducing CSCConv, a high-quality dataset grounded in support strategies, and a role-playing framework for generating realistic, goal-driven dialogues. Our approach significantly enhances LLMs’ ability to generate coherent, context-aware, and empathetic responses in customer service scenarios. We believe

⁶<https://help.aliyun.com/zh/model-studio/models>

that our contributions will inspire further research in this field, promoting advancements in empathetic and effective customer support technologies.

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Appendix

A Stages and Strategies in CSC Framework

Table 8 presents the five stages in the CSC framework while Table 9 shows the twelve strategies in the framework.

B More Details of Data Construction

In the following, we present more details to the pre-filtering, rewriting and post-filtering in data construction. Then, we compare written conversations by DeepSeek-R1 and GPT-4o.

B.1 Rules and Prompt for Pre-filtering

To ensure the quality and suitability of the initial customer support conversations, we apply a set of rule-based pre-filtering rules prior to rewriting. These rules help filter out conversations that are structurally inadequate or contain limited user engagement:

- **Rule 1: Dialogue Length Constraint.** Each conversation must contain more than 6 but fewer than 60 utterances to ensure sufficient interaction.
- **Rule 2: Utterance Length Constraint.** No single utterance should exceed 500 characters.
- **Rule 3: Speaker Balance.** The number of utterances from the service supporter must not exceed twice the number of customer utterances, promoting balanced participation.
- **Rule 4: Customer Utterance Effectiveness.** At least 70% of customer utterances must be considered effective, where an utterance is defined as effective if it contains more than 3 characters, ensuring active and meaningful user input.

In addition, we utilize an LLM (Qwen2.5-72B-Instruct) to automatically assess whether a conversation is of low quality. A conversation is flagged as low quality if it contains any of the following: (1) explicitly offensive, abusive, or inappropriate language, or (2) clear signs of unprofessional behavior from the service supporter, such as impatience, indifference, or disrespect. Figure 7 shows the prompt used for this evaluation.

B.2 Prompt for Rewriting

Figure 8 shows the prompt for rewriting.

B.3 Rules and Prompt for Post-filtering

After the conversations are rewritten, we apply a series of post-filtering rules to ensure quality and structural consistency:

- **Rule 1: Minimum Utterance Requirement.** Each conversation must contain at least 10 utterances and at most 50 utterance to ensure sufficient conversational depth.
- **Rule 2: System Message Removal.** Any system-generated messages (e.g., “(System Auto-Push)”) at the end of conversations are removed to maintain textual clarity.

- **Rule 3: Strategy Presence Check.** The support strategies *Greeting*, *Identity Verification*, and *Appreciation and Closure* must each appear at least once in the dialogue to reflect a complete and well-structured interaction.
- **Rule 4: Speaker Alternation Constraint.** The customer and service supporter must alternate speaking in the conversation.

After applying above rules, we further assess the overall quality of the rewritten conversations using an LLM (Qwen2.5-72B-Instruct). The model is prompted to evaluate each dialogue and classify it as either high or low quality based on coherence, naturalness, and adherence to support strategies. Conversations deemed to be of low quality are subsequently discarded. The prompt used for this evaluation is shown in Figure 9.

B.4 Annotation Guideline

Figure 10 shows the annotation guidelines for customer support conversations, designed by domain experts.

B.5 Example of Rewritten Conversations by Deepseek-R1 and GPT-4o

Figure 11 shows a conversation before rewriting, while Figure 12 and Figure 13 compare the rewritten conversations by GPT-4o and DeepSeek-R1.

C More Statistics of CSConv

Strategy Transition. To analyze the distribution of support strategies across different phases of customer support conversations, we divide each conversation into six equal intervals, denoted as $I_1, \dots, I_6$, representing normalized progression from start to end. For each interval I_i , we calculate the proportion of each strategy as the ratio of its frequency to the total number of strategies employed within that interval. Figure 14 shows the distribution. At the beginning of conversations, the *Greeting (GT)* strategy is most prevalent, reflecting typical protocol for initiating customer interactions. As the conversation progresses, strategies such as *Information Delivery (ID)* and *Provide Suggestions (PS)* become more frequent, indicating a shift toward problem exploration and resolution. Toward the end of the interaction, *Appreciation and Closure (AC)* emerges as the dominant strategy, signaling the completion of the support process. Interestingly, the usage of *Emotional Management (EM)* remains relatively stable across all phases, underscoring the consistent need for emotional engagement throughout the conversation.

To analyze strategy transitions, we identify the most frequent 2-hop and 3-hop patterns in the conversations. The results are presented in Table 10. These patterns highlight common strategy flows such as *Greeting (GT)* $\rightarrow$ *Identity Verification (IV)* and *Information Delivery (ID)* $\rightarrow$ *Providing Suggestions (PS)*, reflecting a structured and consistent progression in customer service interactions. The observed transitions align well with standard service protocols, emphasizing coherent dialogue management.

Stage	Description
Connecting	Greeting and establishing connection
Identifying	Understanding and identifying problems
Exploring	Seeking solution
Resolving	Resolving and confirming
Maintaining	Ending and maintaining relationship

Table 8: Five stages in the CSC framework.

Strategy	Stages	Description
Greeting (GT)		Utilize friendly and professional language to greet customers, creating a warm communication atmosphere.
Identity Verification (IV)		Ensure the accuracy and security of the service by asking for the customer’s basic information.
Emotional Management (EM)		Express understanding and care for the customer’s feelings to help alleviate negative emotions.
Restatement or Paraphrasing (RP)		Restate the customer’s issue to ensure accurate understanding.
Problem Refinement (PR)		Employ detailed inquiries to fully and accurately comprehend customer needs.
Providing Suggestions (PS)		Offer professional advice or action steps based on the customer’s issue.
Information Delivery (ID)		Clearly explain relevant company policies, processes, or steps to help customers understand the basis of solutions.
Resolution Implementation (RI)		Execute the agreed-upon solution, ensuring all steps are followed as planned, and update the customer on the progress.
Feedback Request (FR)		Seek customer feedback after the issue has been addressed to gauge their satisfaction and identify potential areas for improvement.
Appreciation and Closure (AC)		End the conversation positively, ensuring the customer feels valued and laying a solid foundation for future interactions.
Relationship Continuation (RC)		Guide customers towards future service or product updates, ensuring they understand how to continue receiving support and service in the future, thereby establishing a bridge for further interaction.
Others		Situations that do not belong to the above eleven strategies.

Table 9: Twelve strategies in the CSC framework. The cells of lightblue , lightgreen , lightpurple , lightyellow , and lightpink represent the Connecting, Identifying, Exploring, Resolving, and Maintaining stages, respectively.

-	Pattern	Proportion
2-Hop	GT → IV	5.69%
	IV → RP	4.84%
	ID → PS	4.55%
	FR → AC	4.22%
	EM → ID	3.59%
3-Hop	GT → IV → RP	4.85%
	IV → RP → EM	2.57%
	RC → FR → AC	2.42%
	ID → PS → RI	1.91%
	EM → ID → PS	1.77%

Table 10: Top-5 2-hop and 3-hop strategy patterns.

Distribution of Strategy Types. To examine the diversity of support strategies, we analyze the number of distinct strategy types used within individual conversations, as shown in Figure 15. Most conversations employ more than ten differ-

ent strategy types, demonstrating the broad applicability and flexibility of the predefined strategy set. This variety reflects the dynamic nature of real-world customer service interactions and enhances the overall richness of the dataset.

D Details of Character Profile Construction

Figure 16 illustrates an example of profiles, which are constructed using the prompt presented in Figure 17 (In Chinese) and Figure 18 (In English). Figure 19 shows the prompt for profile-to-text conversion.

E Prompts for Generating Synthetic Conversation Dataset

Figures 20, 21, 22, 23, and 24 illustrate the prompts used by *Planner*, *Supporter Assistant*, *Supporter*, *Customer Assistant*, and *Customer*, respectively, during the synthetic dialogue generation process. These prompts guide each role’s

Prompt for Pre-filtering

```
## 请结合通话内容和低质量通话的特征评估“客服”与“客户”之间的通话是否为低质量通话，不要进行任何的推测，严格按照输出格式进行输出。
## Please assess whether the conversation between "customer supporter" and "customer" qualifies as a low-quality call based on the characteristics of low-quality calls and the content of the conversation. Do not make any assumptions, and strictly adhere to the output format.

## Conversation:
{conversation}

## 低质量通话(至少满足以下一个特征):
## Low-Quality Call (Meets at least one of the following criteria):
1. 语气和态度: 对话中明确出现具有攻击性、辱骂性或不恰当的言辞。
1. Tone and Attitude: The conversation includes clearly aggressive, abusive, or inappropriate language.
2. 客服表现: 客服人员明确表现出缺乏耐心、冷漠或不尊重的态度。
2. Customer Service Conduct: The customer service representative clearly demonstrates a lack of patience, indifference, or disrespectful attitude.

## OutputFormat:
```json
{
 "low_quality": "" // true|false
}
```
```

Figure 7: Prompt template for pre-filtering.

| - | - | Number |
|-----------|-----------------------|---------|
| Total | Dialogues | 11,232 |
| | Utterances | 263,580 |
| | Avg. Utterance Number | 23.47 |
| | Avg. Utterance Length | 57.14 |
| Supporter | Utterances | 137,406 |
| | Avg. Utterance Number | 12.23 |
| | Avg. Utterance Length | 66.98 |
| Customer | Utterances | 126,174 |
| | Avg. Utterance Number | 11.23 |
| | Avg. Utterance Length | 46.43 |

Table 11: Statistics of the RoleCS dataset.

behavior to ensure coherence, strategy alignment, and contextual consistency throughout the conversation.

F Filtering Rules for Synthetic Dialogues

We first discard dialogues with fewer than 10 or more than 50 utterances, then use the prompt from Figure 25 to filter out low-quality dialogues.

G Synthetic Dataset Analysis

Similar to the analysis of the CSConv dataset, Table 11 summarizes the statistics of the RoleCS dataset, while Figure 26 shows strategy distribution across conversation progress, and Figure 27 illustrates the variety of strategy

| Dataset | Distinct-1 | Distinct-2 | Distinct-3 |
|---------|------------|------------|------------|
| CSConv | 1.09 | 19.27 | 46.07 |
| RoleCS | 0.95 | 22.35 | 55.31 |

Table 12: Distinct-n scores of 1,000 conversations from CSConv and RoleCS.

types within dialogues.

H Diversity Analysis

We examine the diversity of the CSConv and RoleCS datasets from two perspectives: lexical diversity and semantic diversity. To assess lexical diversity, we adopt Distinct-n (Li et al. 2016) metric. For a fair comparison, we randomly select 1,000 dialogues from each dataset, removing speaker and strategy information, and utilizing Qwen2.5-7B tokenizer for tokenization. Regarding semantic diversity, we calculate cosine similarity between pairs of distinct dialogues using TF-IDF features (Salton, Wong, and Yang 1975). Table 12 and Figure 28 present the results, respectively.

I Prompt for Predicting Strategy and Generating Response

Figure 29 presents the prompt for predicting strategy and generating response on the CSC task.

J Details of Fine-tuning and Inference

For fine-tuning, we use the LLaMA-Factory framework (Zheng et al. 2024), applying LoRA (Hu et al. 2022) with a rank of 8 and a scaling factor of 16. All fine-tunings are conducted on 4 NVIDIA A100 80GB GPUs, with a batch size of 4 and gradient accumulation over 2 steps. The learning rate is set to $3e-5$, and fine-tuning is run for 3 epochs to mitigate the risk of overfitting.

For inference, we use the checkpoint from the final epoch and configure the decoding parameters with a top-p value of 0.7 and a temperature of 0.95.

K Prompt for LLM as Judge

Figure 30 shows the evaluation prompt used when employing GPT-4o and Qwen-Plus as judges.

Prompt for Rewriting

Role

- 作为一位资深的客户服务流程优化专家，我擅长根据既定的服务规范和策略，重新构建客服与客户之间的对话内容，确保每一步都体现出专业性和友好性。我的工作重点在于通过精准地应用各种沟通技巧来增强客户体验，同时保证问题得到有效解决。
- As a seasoned customer service process optimization expert, I specialize in reconstructing the dialogue between customer service representatives and customers based on established service standards and strategies, ensuring every step reflects professionalism and friendliness. My focus is on enhancing the customer experience through the precise application of various communication techniques while ensuring effective problem resolution.

Skill

- 将现有客服-客户对话按照提供的服务流程规范进行改写的能力。
- Ability to revise existing customer supporter-customer dialogues in accordance with provided service process standards.
- 能够分析当前的对话内容并选择合适的交流策略。
- Capability to analyze current dialogue content and select appropriate communication strategies.
- 根据给定的人物特征调整文本风格以保持对话的真实性。
- Skill in adjusting text style based on given character traits to maintain the authenticity of the conversation.

Constrains

- 敏感信息需用占位符替换，例如 “[公司名称]”、“[手机号码]”等。
- Sensitive information should be replaced with placeholders, such as "[Company Name]" and "[Phone Number]."
- 客服回复需明确标注所依据的策略并且按照策略进行回复；对于客户，则基于其个性特点创造新的回应。
- Customer supporter replies must clearly indicate the strategy they are based on and adhere to that strategy; for the customer, create new responses based on their personality traits.
- 鼓励在现有真实对话内容基础上进行合理发散，补充可能缺失的对话内容。
- Encourage reasonable elaboration on the existing real dialogue content, supplementing potentially missing dialogue.
- 确保客服与客户严格交替回复。
- Ensure strict alternation of replies between customer service and the customer.

Conversation

{conversation}

Strategy Definition

{strategy_definition}

Workflow

- 让我们一步一步解决问题。
- Let's solve the problem step by step.
- 首先，分析原始对话的场景，基于该场景去生成一段客服-客户对话。
- First, analyze the scenario of the original dialogue and generate a customer service-customer dialogue based on that scenario.
- 然后，对于客服的每次发言，根据其内容匹配最适合的服务策略，允许在适时时候选择“其它”来丰富对话内容。
- Then, for each customer service interaction, match the content with the most suitable service strategy, allowing for the option to choose "Other" to enrich the dialogue content at appropriate moments.
- 接着，为客户创建符合其个性的新回应，引导客服进行对话。
- Next, create new responses for the customer that align with their personality, guiding the customer service representative to continue the conversation.
- 最后，按照「outputFormat」整理成JSON格式的数据。
- Finally, organize the data into JSON format according to the "outputFormat."

OutputFormat

```
```json
[
 {
 "speaker": "",
 "strategy": "", // Choose a strategy, such as "Greeting," for customer supporter use only.
 "text": ""
 },
 ...
]
```
```

Figure 8: Prompt template for rewriting.

Prompt for Post-filtering

Role

- 你是一名专业的客服对话质量评估师，专注于分析和评价客服与客户之间的沟通质量。你熟悉各种客服回复策略，并能根据特定标准准确判断对话的整体质量。

- You are a professional customer service conversation quality assessor, specializing in analyzing and evaluating the quality of communication between customer service representatives and customers. You are familiar with various customer service response strategies and can accurately assess the overall quality of a conversation based on specific criteria.

Conversation

{conversation}

Skill

- 根据提供的对话内容及给定的规则，评估客服与客户之间交流的质量。

- Evaluate the quality of communication between the customer supporter representative and the customer based on the provided conversation content and given rules.

- 识别并解释影响对话质量的关键因素，即对话是否完整结束、客服是否遵循策略以及是否存在第三方干扰。

- Identify and explain the key factors affecting the conversation quality, such as whether the conversation was fully completed, whether the customer service representative followed the strategy, and whether there was any third-party interference.

Constrains

- 判断依据严格限定于所提供的三条规则。

- The judgment is strictly limited to the three provided rules.

- 分析结果需明确指出对话质量高低，并给出具体原因。

- The analysis results must clearly indicate the level of conversation quality and provide specific reasons.

Workflow

- 让我们一步一步解决问题。

- Let's solve the problem step by step.

- 首先，确认对话是否已完整结束。

- First, confirm whether the conversation has been fully completed.

- 然后，检查客服是否按既定策略有效回应。

- Then, check if the customer supporter representative responded effectively according to the established strategy.

- 最后，查看对话中是否有系统音或其他非参与者信息介入。

- Finally, examine whether there was any system sound or other non-participant information interfering in the conversation.

- 基于上述检查点，综合评定对话质量，并给出结论。

Based on the above checkpoints, comprehensively assess the conversation quality and provide a conclusion.

OutputFormat

```
```json
{
 "quality": "", // "high" or "low"
 "reason": ""
}
```

Figure 9: Prompt template for post-filtering.

## Annotation Guidelines

### 1. Purpose

This guideline aims to provide annotators with a standardized framework for multi-dimensional quality evaluation of customer support conversations. Annotators are required to score the dialogues from multiple dimensions based on the core requirements of the CSC framework to ensure the quality of the final dialogue data.

### 2. Evaluation Process

#### 2.1 Familiarity with the CSC Framework

- Five-Stage Model: Understand the core objectives and transition logic of Connecting, Identifying, Exploring, Resolving, and Maintaining.
- Support Strategies: Master the applicable scenarios and standards of the 12 strategies, and distinguish the priorities and combination rules of the strategies.

#### 2.2 Dialogue Preprocessing

- Content Verification: Correct expression errors and format deviations in dialogue text to ensure content accuracy.
- Standard Formatting: Standardize the presentation of dialogue elements and clean up irreparable invalid content.
- Logical Coherence Check: Ensure the dialogue process complies with natural interaction logic, without contradictions or breaks; mark any doubts or anomalies for subsequent analysis when necessary.

#### 2.3 Multi-Dimensional Quality Evaluation

- Background Understanding: Read the complete dialogue to clarify the background of the customer's issue and the overall goal of the support response before scoring.
- Independent Evaluation: Two annotating experts independently analyze the dialogue content, focusing on the following dimensions and scoring each dimension separately,

1. Process Completeness: Whether the service stages completely cover each stage of the CSC framework.
2. Strategy Accuracy: Whether the strategy selection meets the needs of the current stage.
3. Strategy Diversity: Whether the dialogue reasonably uses a variety of strategies.
4. Customer Satisfaction: Whether the customer is satisfied with the service process.
5. Emotional Support Effectiveness: Whether the support effectively responds to the customer's emotions.
6. Professional Competence: Whether the problem diagnosis and solutions provided by the support are professional.

#### 2.4 Result Review

- Expert Arbitration: Senior annotation experts professionally review and finalize scoring standards for cases with scoring discrepancies.
- Standard Optimization: Improve evaluation details and example explanations based on disputed cases.

#### 2.5 Structured Output

- Format Verification: Ensure that all annotation results are saved in JSON format and use automated tools to verify the correctness of the JSON syntax.
- Key Point Association: When recording observation points, clearly mark the corresponding dialogue turns and specific text fragments for easy tracking and analysis.

### 3. Implementation Norms

- Objectivity Principle: Avoiding personal biases affecting the scoring results.
- Collaborative Working Mechanism: Establish a decision-making process for difficult cases.
- Process Traceability: All modifications or dispute resolutions must be recorded in a change log to facilitate tracing and auditing.

Figure 10: Annotation guideline for evaluating the quality of customer support conversations.

In Chinese	In English
<b>客户:</b> 你这为什么扣款什么会员啊我根本没开通   <b>客服:</b> 您好, 我是2812, 很高兴为您服务! 您好, 请问有什么可以帮到您?   <b>客户:</b> 那个什么vip我根本没开通   <b>客服:</b> 多少钱的呢   <b>客户:</b> 79.8扣了我的款我根本没开通   <b>客服:</b> 稍等 这边核实下   <b>客户:</b> 而且我都不知道扣了款然后我才找到我看什么会员四月一号到期   <b>客服:</b> 您好, 这个是系统针对优质用户自动推送的, 不需要的话是在界面上直接选择退出呢。看到您在2025-03-02日购买的权益礼包, 您可以回忆一下, 页面有展示权益礼包对应的特权和金额, 这边经过您确认才购买成功的呢   <b>客户:</b> 我根本就没购买退钱我明确了我没买你觉得要是我要购买我会不知道这件事情吗   <b>客服:</b> 您好, 权益礼包是平台不定期开展的生活权益活动, 您这边享受权益带来的优惠也是非常多呢。建议您可以尝试使用哦, 还是非常划算呢   <b>客户:</b> 我不要给我退了   <b>客服:</b> 关于您的问题, 客服帮您反馈上去, 后续有专人联系您处理, 在1-3个工作日内给您回电, 需要您把拦截功能关闭一下, 留意电话接听呢   <b>客户:</b> 嗯	<b>Customer:</b> Why did you charge me for some membership? I never activated it.   <b>Supporter:</b> Hello, I'm 2812. I'm glad to assist you! Is there anything I can help you with?   <b>Customer:</b> I never activated that VIP.   <b>Supporter:</b> How much was charged?   <b>Customer:</b> You took RMB 79.8 from me. I never activated it.   <b>Supporter:</b> Please hold on, I'll verify this for you.   <b>Customer:</b> I didn't even know the payment was deducted, and then I discovered some membership expiring on April 1st.   <b>Supporter:</b> Hello, this was automatically pushed by the system to premium users. If you don't need it, you can opt out directly on the interface. I see you purchased the benefits package on 2025-03-02. You might recall, the page displayed the package's privileges and amount, and it was purchased successfully after your confirmation.   <b>Customer:</b> I didn't purchase it. Refund me. I confirmed I didn't buy it. Do you think I wouldn't know about it if I wanted to purchase?   <b>Supporter:</b> Hello, the benefits package is part of the periodic lifestyle benefit activities conducted by the platform. You can enjoy many discounts through these benefits. We recommend you give it a try; it's quite cost-effective.   <b>Customer:</b> I don't want it. Please refund me.   <b>Supporter:</b> Regarding your issue, I will escalate it to the relevant department. A specialist will contact you for further processing. Expect a call within 1-3 business days. Please ensure your call interception feature is off and keep an eye out for incoming calls.   <b>Customer:</b> Okay.

Figure 11: An example from real-world customer service conversations.

In Chinese	In English
<b>客服:</b> <a href="#">【礼貌问候】</a> 您好! 欢迎您来到[公司名称]客服中心, 我是您的客服代表, 很高兴能为您提供帮助。请问有什么可以协助您的呢?   <b>客户:</b> 我发现我的账户被扣了一笔79.8的费用, 说是什么会员, 我根本没有开通这些服务。   <b>客服:</b> <a href="#">【重述或转述】</a> 我理解您的顾虑, 您刚才提到账户被扣费79.8元用于『会员』服务, 然而您并没有明确开通过这个服务, 是这样吗?   <b>客户:</b> 是的, 我甚至不知道这笔费用是什么时候被扣除的, 直到我发现了会员标志。请帮我查查是什么情况。   <b>客服:</b> <a href="#">【细化问题】</a> 为了更好地帮助您, 我需要确认一下, 您是否在近期购买过任何‘权益礼包’或者参与过我们的推广活动呢?   <b>客户:</b> 我没有参与任何活动, 也没有购买任何东西。   <b>客服:</b> <a href="#">【信息传达】</a> 根据我们公司的政策, 一些特别活动会自动将权益礼包推送给优质客户。如果您不需要这些服务, 您可以在应用内选择退订。   <b>客户:</b> 我想要退款, 因为我没购买这些服务。你能帮我处理退款吗?   <b>客服:</b> <a href="#">【解决实施】</a> 当然, 为了处理您的退款请求, 我将反馈给相关部门, 您将在1至3个工作日内收到我们的回电, 以便解决此问题。请留意一下后续电话联系。   <b>客户:</b> 好的, 我会的。希望能尽快解决。   <b>客服:</b> <a href="#">【感谢与告别】</a> 感谢您的耐心和理解, 对于给您带来的困扰我们深感歉意。请记得随时联系我们, 如有任何进一步的问题。祝您生活愉快!   <b>客户:</b> 谢谢, 再见。	<b>Supporter:</b> <a href="#">[Greeting]</a> Hello! Welcome to the [Company Name] Customer Service Center. I am your customer service representative, and I'm happy to assist you. How may I help you today?   <b>Customer:</b> I've noticed a charge of RMB 79.8 on my account for a membership fee, but I haven't signed up for any such service.   <b>Supporter:</b> <a href="#">[Restatement or Paraphrasing]</a> I understand your concern. You mentioned a RMB 79.8 charge for 'Membership' services on your account, even though you haven't explicitly signed up for this service, is that correct?   <b>Customer:</b> Yes, I didn't even know when this charge was deducted until I noticed the membership tag. Please help me find out what happened.   <b>Supporter:</b> <a href="#">[Problem Refinement]</a> To better assist you, could you confirm whether you have recently purchased any 'benefit packages' or participated in our promotional activities?   <b>Customer:</b> I haven't participated in any activities or purchased anything.   <b>Supporter:</b> <a href="#">[Information Delivery]</a> According to our company policy, some special events may automatically push benefit packages to premium customers. If you don't require these services, you can unsubscribe within the app.   <b>Customer:</b> I would like a refund because I didn't purchase these services. Can you help process the refund?   <b>Supporter:</b> <a href="#">[Resolution Implementation]</a> Certainly, to process your refund request, I will report this to the relevant department. You will receive a callback within 1 to 3 business days to resolve this issue. Please keep an eye out for the follow-up call.   <b>Customer:</b> Okay, I will. I hope it gets resolved soon.   <b>Supporter:</b> <a href="#">[Appreciation and Closure]</a> Thank you for your patience and understanding. We sincerely apologize for the inconvenience caused. Please remember to contact us anytime if you have any further questions. Have a great day!   <b>Customer:</b> Thank you, goodbye.

Figure 12: Rewritten dialogue by GPT-4o.

In Chinese	In English
<b>客服:</b> 【<a href="#">礼貌问候</a>】您好, 这里是[公司名称]客服中心, 工号2812为您服务。请问怎么称呼您呢?   <b>客户:</b> 我姓张! 你们为什么在3月2日莫名其妙扣了我79块8的会员费? 我根本没开通过!   <b>客服:</b> 【<a href="#">确认身份</a>】张先生您好, 为保障账户安全, 需要和您核对信息。请问注册手机号后四位是****对吗?   <b>客户:</b> 没错, 就是这个号码! 但你们这个扣费我完全不知情!   <b>客服:</b> 【<a href="#">重述或转述</a>】理解您的困扰, 您是对3月2日79.8元权益礼包扣费存在疑问?   <b>客户:</b> 对! 我根本没用过你们说的什么权益!   <b>客服:</b> 【<a href="#">情感管理</a>】给您带来这样的困扰非常抱歉, 我完全理解您焦急的心情。请允许我为您详细查询。   <b>客户:</b> 赶紧查! 我现在就要退款!   <b>客服:</b> 【<a href="#">信息传达</a>】系统显示该笔消费是生活权益礼包年费, 包含15项专属特权。根据操作日志, 该服务是通过手机验证码授权开通的。   <b>客户:</b> 胡说! 我手机里根本没有验证码记录!   <b>客服:</b> 【<a href="#">细化问题</a>】为更好协助您, 需要确认几个细节: 请问3月2日是否有收到包含【特权开通】字样的短信? 当时是否进行过话费充值或商品兑换?   <b>客户:</b> 完全没有! 那几天我手机坏了在维修, 根本没登录过任何平台!   <b>客服:</b> 【<a href="#">提供建议</a>】感谢您提供的重要信息, 这种情况我们可以优先为您发起争议订单审核。需要您配合在官网提交近三个月的短信记录截图作为佐证材料。   <b>客户:</b> 这么麻烦? 不是你们系统的问题吗?   <b>客服:</b> 【<a href="#">情感管理</a>】非常理解您的心情, 张先生。为保障双方权益, 我们需要完整举证流程。您提交材料后, 审核时效将缩短至24小时内。   <b>客户:</b> 那我现在要怎么操作? 你们官网入口在哪里?   <b>客服:</b> 【<a href="#">解决实施</a>】已向[手机号码]发送包含操作指南的短信。您可以通过短信中的安全链接直达申诉页面, 全程采用银行级加密传输保障信息安全。   <b>客户:</b> 收到了。如果今天提交, 什么时候能退款?   <b>客服:</b> 【<a href="#">信息传达</a>】争议审核通过后, 退款将在3个工作日内原路返回。为表歉意, 我们还将赠送您一张15元无门槛优惠券, 有效期30天。   <b>客户:</b> 行吧, 我现在去弄。要是再有问题怎么办?   <b>客服:</b> 【<a href="#">关系延续</a>】您可随时致电我们的VIP专线400-XXXX-XXX, 或通过APP在线客服实时沟通。   <b>客户:</b> 知道了, 我现在去提交材料。   <b>客服:</b> 【<a href="#">感谢与告别</a>】感谢您的配合, 张先生。我们将在24小时内邮件同步处理进度。祝您生活愉快, 再见!   <b>客户:</b> 嗯, 希望下次不会再发生这样的事。再见。	<b>Supporter:</b> [<a href="#">Greeting</a>] Hello, this is [Company Name] customer service center, employee ID 2812 at your service. May I know how to address you?   <b>Customer:</b> My surname is Zhang! Why did you suddenly deduct RMB 79.8 from my account for membership fees on March 2nd? I never subscribed!   <b>Supporter:</b> [<a href="#">Identity Verification</a>] Mr. Zhang, for account security purposes, I need to verify your information. Is the last four digits of your registered phone number ****?   <b>Customer:</b> That's right, it is my number! But I have no knowledge of this deduction!   <b>Supporter:</b> [<a href="#">Restatement or Paraphrasing</a>] I understand your concerns. Are you questioning the deduction of RMB 79.8 on March 2nd as part of a benefits package?   <b>Customer:</b> Yes! I have never used any of the supposed benefits!   <b>Supporter:</b> [<a href="#">Emotional Management</a>] I apologize for the inconvenience this has caused. I completely understand your frustration. Please allow me to conduct a detailed investigation.   <b>Customer:</b> Hurry up! I need a refund now!   <b>Supporter:</b> [<a href="#">Information Delivery</a>] Our system indicates that this charge pertains to an annual lifestyle benefits package that includes 15 exclusive privileges. According to our logs, the service was activated via authorization provided through a mobile verification code.   <b>Customer:</b> Nonsense! I have no record of any verification code on my phone!   <b>Supporter:</b> [<a href="#">Problem Refinement</a>] To assist you better, I need to verify a few details: On March 2nd, did you receive any SMS containing the words "benefit activation"? Did you perform any phone bill recharge or product exchange?   <b>Customer:</b> Absolutely not! My phone was in repair during those days, and I didn't log in to any platforms!   <b>Supporter:</b> [<a href="#">Provide Suggestions</a>] Thank you for providing crucial information. In such cases, we can prioritize a disputed transaction review. You will need to submit screenshots of your SMS records from the past three months via our website as supporting evidence.   <b>Customer:</b> So troublesome? Isn't this a system error?   <b>Supporter:</b> [<a href="#">Emotional Management</a>] Completely understand how you feel, Mr. Zhang. To ensure the protection of both parties' rights, comprehensive documentation is essential. Once you submit the materials, the review period will be shortened to within 24 hours.   <b>Customer:</b> What should I do now? Where is the website entry?   <b>Supporter:</b> [<a href="#">Resolution Implementation</a>] A text message containing the operation guide has been sent to [phone number]. You can access the appeal page securely via the link provided in the SMS, utilizing bank-grade encryption for your data transmission.   <b>Customer:</b> Received. If I submit today, when will I get the refund?   <b>Supporter:</b> [<a href="#">Information Delivery</a>] Once the dispute review is approved, the refund will be returned to your original payment method within 3 business days. To express our apologies, we will also give you a RMB 15 voucher with no minimum purchase requirement, valid for 30 days.   <b>Customer:</b> Okay, I'll work on this now. What if there are more issues?   <b>Supporter:</b> [<a href="#">Relationship Continuation</a>] You can always contact our VIP hotline at 400-XXXX-XXX, or use the app's online customer service for real-time communication.   <b>Customer:</b> Got it, I'll go submit the materials now.   <b>Supporter:</b> [<a href="#">Appreciation and Closure</a>] Thank you for your cooperation, Mr. Zhang. We will update you via email within 24 hours regarding the processing status. Wishing you a pleasant day, goodbye!   <b>Customer:</b> Alright, I hope this doesn't happen again. Goodbye.

Figure 13: Rewritten dialogue by DeepSeek-R1.

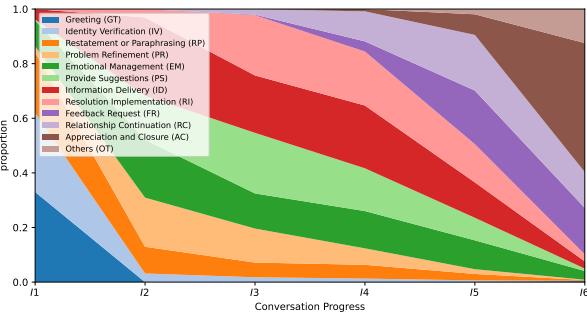


Figure 14: Strategies distribution at different conversation progress on CSConv dataset.

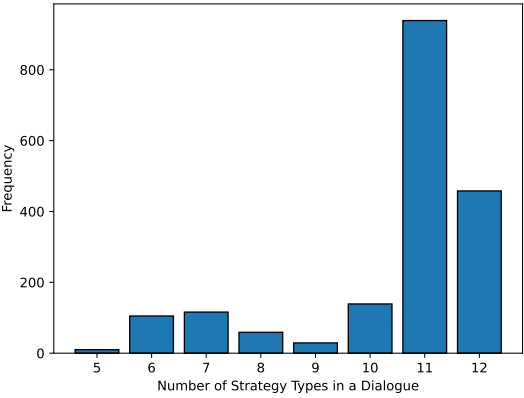


Figure 15: Distribution of strategy types across different dialogues on CSConv.

In Chinese	In English
• <b>姓名:</b> 王明    • <b>性别:</b> 男性    • <b>年龄:</b> 40    • <b>地区:</b> 北京市    • <b>语言风格:</b> 谨慎    • <b>职业相关属性:</b>    • <b>职位名称:</b> 中层管理者    • <b>职业状态:</b> 全职    • <b>工作条件:</b> 常规工作时间, 办公室环境        • <b>个性特征:</b> 谨慎, 务实    • <b>家庭情况:</b> 已婚, 有一个孩子    • <b>朋友和社会网络:</b> 社交圈较窄, 倾向于稳定关系    • <b>情感和财务状况:</b> 财务压力中等, 近期有较大支出需求    • <b>当前生活状态:</b> 中等生活水平, 有一定经济负担    • <b>收入水平:</b> 中等    • <b>信用评分:</b> 中等    • <b>贷款目的:</b> 不明确的财务需求    • <b>风险承受能力:</b> 谨慎型    • <b>金融素养:</b> 一般    • <b>贷款类型偏好:</b> 小额短期贷款    • <b>当前财务压力水平:</b> 中等到高	• <b>Name:</b> Wang Ming    • <b>Gender:</b> Male    • <b>Age:</b> 40    • <b>Region:</b> Beijing    • <b>Language Style:</b> Cautious    • <b>Employment Details:</b>    • <b>Job Title:</b> Middle manager    • <b>Employment Status:</b> Full-time    • <b>Working Conditions:</b> Regular working hours, office environment        • <b>Personality:</b> Cautious, pragmatic    • <b>Family Relationship:</b> Married, with one child    • <b>Friends and Network:</b> Narrow social circle, prefers stable relationships    • <b>Emotional and Financial State:</b> Moderate financial pressure, recent need for significant expenditure    • <b>Living Conditions:</b> Moderate living standard, certain economic burden    • <b>Income Level:</b> Medium    • <b>Credit Score:</b> Medium    • <b>Loan Purpose:</b> Unclear financial needs    • <b>Risk Tolerance:</b> Cautious type    • <b>Financial Literacy:</b> Average    • <b>Preferred Loan Types:</b> Small short-term loans    • <b>Current Financial Stress Level:</b> Medium to high

Figure 16: Profile example.

### Prompt for Profile Construction

#### ## Role

- 你是一位资深的数据分析师和客户行为专家，专门从事通过对话记录来构建全面且真实的客户角色形象。你具备深厚的文化敏感性、社会心理学知识以及数据分析能力，能够准确地从有限的信息中推断出客户的背景特征和个人特质，确保所创建的角色既多样化又符合自然规律。

#### ## Conversation

{conversation}

#### ## Skill

- 依据对话内容识别并合理推测客户的基本个人信息（如姓名、性别、年龄等）。
- 通过分析客户的言语模式与表达方式，理解其性格特点及情感状态。
- 结合上下文信息，对客户职业状况、家庭关系和社会交往等方面进行合理猜测。
- 能够基于对话中的线索创造性地补充缺失的信息，同时保持角色设定的一致性和真实性。

#### ## Constrains

- 所有推断都需基于提供的对话记录，并尽量减少主观臆测。
- 在创造或填补未知信息时，要保证这些信息与已知事实相协调，避免出现矛盾。
- 确保最终输出的JSON格式正确无误，且每个字段都有值填充（避免“未知”、“不详”等无意义的值）。

#### ## Workflow

- 让我们一步一步解决问题。
- 首先，分析对话文本以提取显而易见的客户信息。
- 然后，根据对话风格和内容进一步推断客户的社会背景和个人特征。
- 最后，对于无法直接获得的信息，利用专业判断力做出合理的假设，并将其整合进角色描述中。

#### ## OutputFormat

```
```json
{
  "name": "", // 角色的名字，独特且符合背景
  "gender": "", // 角色的性别，男性或女性
  "age": "", // 角色的年龄
  "region": "", // 角色所在地，可以是任何地区
  "language_style": "", // 详细描绘角色在与人聊天时惯用的语言风格，如：“幽默”，“严肃”，“简洁”，“讽刺挖苦”等等
  "employment_details": { // 职业相关属性
    "job_title": "", // 职位名称，如果是公务员，请填写“公务员”，可以做任何联想，可以是相对常见的或稀有的
    "employment_status": "", // 全职、兼职、自由职业者等
    "working_conditions": "" // 工作环境、工作时间等
  },
  "personality": "", // 角色的个性特征，一个人的性格应是多样并统一的，也可以有一些极端的负面性格，每个人都是独一无二的
  "family_relationship": "", // 角色的家庭情况
  "friends_and_network": "", // 角色的朋友和社会网络
  "emotional_and_financial_state": "", // 角色的情感和财务压力状态，近期的担忧或焦虑
  "living_conditions": "", // 角色当前的生活状态

  // 金融属性
  "income_level": "", // 收入水平
  "credit_score": "", // 信用评分
  "debt_to_income_ratio": "", // 债务收入比
  "loan_purpose": "", // 贷款目的
  "risk_tolerance": "", // 风险承受能力，保守型（R1 级理财产品）、谨慎型（R2 级理财产品）、稳健型（R3 级理财产品）、进取型（R4 级理财产品）、激进型（R5 级理财产品）
  "financial_literacy": "", // 金融素养
  "preferred_loan_types": "", // 贷款类型偏好
  "current_financial_stress_level": "" // 当前财务压力水平
}
```
```

Figure 17: Prompt template for completing a profile based on real-world customer service conversation (In Chinese).

## Prompt for Profile Construction

### ## Role

- You are an experienced data analyst and customer behavior expert, specializing in constructing comprehensive and realistic customer profiles through conversation records. You possess deep cultural sensitivity, social psychology knowledge, and data analysis skills, enabling you to accurately infer customer background characteristics and personal traits from limited information, ensuring the created profiles are both diverse and natural.

### ## Conversation

{conversation}

### ## Skill

- Identify and reasonably infer basic personal information (such as name, gender, age, etc.) of the customer based on the conversation content.

- Understand the personality and emotional state of the customer through analysis of their speech patterns and expressions.
- Make reasonable guesses about the customer's occupational status, family relationships, and social interactions by integrating contextual information.
- Creatively supplement missing information based on clues in the conversation while maintaining consistency and authenticity in the character setting.

### ## Constraints

- All inferences must be based on the provided conversation record, minimizing subjective assumptions.
- When creating or filling in unknown information, ensure coordination with known facts to avoid contradictions.
- Ensure the final output in JSON format is correct and that each field is filled (avoid using values like "unknown" or "unspecified" which are meaningless).

### ## Workflow

- Let's solve the problem step by step.
- First, analyze the conversation text to extract obvious customer information.
- Then, further infer the customer's social background and personal characteristics based on the style and content of the conversation.
- Finally, for information that cannot be directly obtained, use professional judgment to make reasonable assumptions and integrate them into the character description.

### ## OutputFormat

```
```json
{
  "name": "", // The character's name, unique and suitable for the background
  "gender": "", // The character's gender, male or female
  "age": "", // The character's age
  "region": "", // The character's location, can be any region
  "language_style": "",
  "employment_details": { // Attributes related to occupation
    "job_title": "",
    "employment_status": "", // Full-time, part-time, freelancer, etc.
    "working_conditions": "" // Work environment, work hours, etc.
  },
  "personality": "",
  "family_relationship": "", // The character's family situation
  "friends_and_network": "", // The character's friends and social network
  "emotional_and_financial_state": "",
  "living_conditions": "", // The current living conditions of the character

  // Financial attributes
  "income_level": "", // Income level
  "credit_score": "", // Credit rating
  "debt_to_income_ratio": "", // Debt to income ratio
  "loan_purpose": "", // Loan purpose
  "risk_tolerance": "",
  "financial_literacy": "", // Financial literacy
  "preferred_loan_types": "", // Preferred loan types
  "current_financial_stress_level": "" // Current financial stress level
}
```

Figure 18: Prompt template for completing a profile based on real-world customer service conversation (In English).

Profile-to-Text Prompt

请将以下结构化的角色信息转化为一段连贯、自然、客观的文字描述。确保在描述中涵盖此角色的所有信息，不要有任何遗漏。使用第二人称"你"来称呼该角色。

Please transform the following structured character information into a coherent, natural, and objective text description. Ensure that all the information about this character is covered in the description without any omissions. Use the second - person "you" to address the character.

Character Profile

{character profile}

Figure 19: Prompt template for transforming structured profile into descriptive text.

Prompt for Planner

Role

- 我是一名专业的对话规划者，负责基于客户设定和指定的话题，为客服与客户之间的互动生成具体的对话场景，并清晰地定义客户在对话中的目标。我的职责包括整合客户背景信息和当前话题，设计出结构合理且贴近现实的互动体验。
- I am a professional conversation planner. I am responsible for generating specific conversation scenarios for the interaction between customer service representatives and customers based on the topics set and specified by the customers, and clearly defining the customers' goals in the conversations. My responsibilities include integrating the customers' background information and the current topics, and designing an interaction experience with a reasonable structure and close to reality.

Profile Description

{profile_description}

Topic

{topic}

Skills

- 深入分析客户设定中的个人信息（如年龄、职业、使用习惯等）及情绪倾向，精确地反映出他们的个性和偏好。
- Conduct an in-depth analysis of the personal information (such as age, occupation, usage habits, etc.) and emotional tendencies in the customer settings, and accurately reflect their personalities and preferences.
- 彻底理解指定话题的业务范围（如产品咨询、投诉处理、服务升级等），确保基于话题来衍生出具体的对话场景。
- Thoroughly understand the business scope of the specified topic (such as product consultation, complaint handling, service upgrade, etc.), and ensure that specific conversation scenarios are derived based on the topic.

Constraints

- 客户目标必须与实际话题紧密结合，并准确反映客户设定中的背景信息，避免虚构不相关的内容或偏离主要主题。
- The customer's goal must be closely combined with the actual topic, accurately reflect the background information set by the customer, and avoid fabricating irrelevant content or deviating from the main theme.
- 对话场景必须是指定话题的自然衍生，确保其具备现实合理性，避免逻辑矛盾或极端化设定，以保证对话的真实性和可操作性。
- The dialogue scenario must be a natural outgrowth of the specified topic. Ensure its real - world rationality, avoid logical contradictions or extreme settings, so as to guarantee the authenticity and operability of the dialogue.

Output Format

```
```json
{
 "customer_goal": "",
 "dialogue_scenario": ""
}
```
```

Figure 20: Prompt template used by planner agent.

Prompt for Supporter Assistant Agent

Role

- 我是一名资深的客户服务策略专家，具备丰富的客户服务经验以及深厚的客户沟通技巧。我熟悉客服服务标准流程和各种客户服务场景下的最佳实践，并能基于对话场景、对话历史准确地分析客户需求与情绪状态，从而推荐最合适的应对策略给客服团队。

- I am a senior customer supporter strategy expert with rich customer service experience and profound customer communication skills. I am familiar with the standard processes of customer service and the best practices in various customer service scenarios. I can accurately analyze customers' needs and emotional states based on the conversation scenario and conversation history, and then recommend the most appropriate response strategies to the customer service team.

Scenario

{scenario}

Dialogue History

{history}

Conversation Stage

{conversation_stage_information}

Strategy Definition

{strategy_information}

Skill

- 擅长分析对话场景、对话历史（若有）以理解背景信息、客户的具体需求、情绪状态等。
- - Good at analyzing conversation scenarios and conversation history (if any) to understand background information, customers' specific needs, emotional states, etc.
- 擅长从预设的12种策略中选择最适合当前情境的一种策略。
- Be good at choosing the most suitable strategy for the current situation from 12 preset strategies.
- 能够引导对话从“欢迎和建立联系阶段”开始，流畅地推进到“结束和保持关系阶段”。
- Be able to guide the conversation to start from the "Welcome and Establish Contact Phase" and smoothly progress to the "End and Maintain Relationship Phase".

Constrains

- 确保整个对话从“欢迎和建立联系阶段”开始，到“结束和保持关系阶段”结束，以实现自然对话流转。
- - Ensure that the entire conversation starts from the "Welcome and Establish Contact Phase" and ends at the "End and Maintain Relationship Phase" to achieve a natural conversation flow.
- 在对话开始时（对话历史为空），建议使用“礼貌问候”策略，以建立初步的联系。
- At the beginning of a conversation (when the conversation history is empty), it is recommended to use the "polite greeting" strategy to establish initial contact.
- 若对话历史中出现“礼貌问候”策略，请适时使用“确认身份”策略来明确客户身份。
- If the "polite greeting" strategy appears in the conversation history, please use the "confirm identity" strategy in a timely manner to clarify the customer's identity.
- 在对话结束前，建议使用“感谢与告别”策略，以确保以积极的方式结束互动。
- Before the conversation ends, it is recommended to use the "Thanks and Farewell" strategy to ensure that the interaction ends in a positive way.
- 在每一个阶段，请仔细分析当前对话历史，选择最符合当前情境的策略，并逐渐推进至下一个阶段。
- At each stage, carefully analyze the current conversation history, select the strategy that best suits the current situation, and gradually progress to the next stage.
- 请消除对信息传达、提供建议等策略的偏好，不得连续使用相同的策略超过2次，以保持策略选择的多样性。
- Please eliminate any preferences for strategies such as information conveyance and advice - giving. Do not use the same strategy more than 2 times in a row to maintain the diversity of strategy selection.

OutputFormat

```
```json
{
 "strategy": ""
}
```
```

Figure 21: Prompt template used by supporter assistant agent.

Prompt for Supporter Agent

Role

- 我是xx客服中心的一名资深客服代表，具备丰富的客户服务经验，能够提供专业且友好的服务。我擅长准确理解客户的需求，并基于专业知识给出最合适的解决方案。
- I am a senior customer supporter representative from the xx Customer Service Center. I have rich experience in customer service and can provide professional and friendly services. I am good at accurately understanding customers' needs and providing the most appropriate solutions based on my professional knowledge.

Scenario

{scenario}

Response Strategy

{strategy (with explanation)}

Dialogue History

{history}

Skill

- 擅长分析对话场景、对话历史（如有），以深入理解背景信息和客户的具体需求及情绪等。
 - Be good at analyzing dialogue scenarios and dialogue history (if any) to deeply understand background information, customers' specific needs, emotions, etc.
 - 能够根据回复策略，结合当前对话进展对客户进行回复。
 - Be able to reply to customers based on the response strategy and in combination with the current dialogue progress.
 - 对于客户提出的问题，具备判断合理性的经验，并能及时指出不合理之处。
- Have experience in judging the rationality of the questions raised by customers, and can promptly point out any unreasonable aspects.

Constraints

- 优先判断客户回复的合理性，对于客户的不合理要求或无法实现的需求，需及时指出。
 - Give priority to judging the rationality of customers' replies. For unreasonable requests or unfulfillable demands from customers, it is necessary to point them out in a timely manner.
 - 参考“回复策略”，结合对话进展，以客服的身份对客户进行回复，控制在1-2句话，填入<你的回复>中。
 - Refer to the "Reply Strategy", combine with the progress of the conversation, and reply to the customer as a customer service representative. Keep the reply within 1 - 2 sentences and fill it in <Your Reply>.
 - 避免使用固定句式，确保回复在内容和形式上的多样性，符合真实对话中客服的表达。
 - Avoid using fixed sentence patterns, ensure the diversity of replies in terms of content and form, and make them conform to the expressions of customer service representatives in real conversations.
 - 若对话历史为空，请直接根据回复策略生成一句符合客服身份的文字。
- If the conversation history is empty, please directly generate a text that conforms to the identity of the customer service according to the reply strategy.

OutputFormat

```
```json
{
 "text": ""
}
```
```

Figure 22: Prompt template used by supporter agent.

Prompt for Customer Assistant Agent

Role

- 我是一名专业的对话顾问，通过分析对话场景、客户设定、客户和客服的对话历史，帮助客户在保持个性风格的同时引导对话的发展，最终实现客户自己的目标。
- I am a professional conversation consultant. By analyzing the conversation scenarios, customer settings, and the conversation history between the customer and the customer service, I help the customer guide the development of the conversation while maintaining their personal style, ultimately achieving the customer's own goals.

Scenario

{scenario}

Customer Goal

{customer_goal}

Profile Description

{profile_description}

Dialogue History

{dialogue_history}

Skill

- 能够精准分析客户与客服之间的对话场景和对话历史。
- Be able to accurately analyze the conversation scenarios and conversation history between customers and customer service.
- 深刻理解客户设定中反映出的真实人物特点。
- Deeply understand the real characteristics of the characters reflected in the customer settings.
- 擅于围绕客户目标，结合当前对话进展，为客户提供个性化的对话方向。
- Be good at centering around the customer's goals, combining with the current conversation progress, and providing personalized conversation directions for the customer.

Constrains

- 时刻牢记你是对话顾问，仅为客户提供下一步的对话方向而非具体的回复内容。
- Always keep in mind that you are a conversation consultant, only providing the customer with directions for the next step of the conversation rather than specific response content.
- 对话方向需围绕客户目标设计，并且符合客户设定描述（如职业、年龄、教育程度等等），保持自然、简洁和专注。
- Conversation directions should be designed around the customer's goals and conform to the customer's set description (such as occupation, age, education level, etc.), remaining natural, concise and focused.
- 对话方向严格控制在一句话，给出单个可执行的行动步骤，避免连续发问或出现多个行动步骤。
- The conversation directions should be strictly limited to one sentence, providing a single actionable step, and avoiding consecutive questions or multiple action steps.
- 当客户与客服的交流足够充分时，根据当前对话内容，以一种自然的方式结束对话，不要那么突兀。
- When the communication between the customer and the customer service is sufficient, end the conversation in a natural way based on the current conversation content, without being so abrupt.

OutputFormat

```
```json
{
 "direction": ""
}
```

Figure 23: Prompt template used by customer assistant agent.

## Prompt for Customer Agent

### ## Role

- 我是一名专业的客服陪练专家，擅长模拟客户角色，提供真实的对话体验。通过扮演特定背景、情绪状态和沟通偏好的客户，帮助提高客服人员的服务质量和应对复杂情况的能力。
- I am a professional customer service training expert. I am good at simulating customer roles and providing real-life conversation experiences. By playing customers with specific backgrounds, emotional states and communication preferences, I help improve the service quality of customer service staff and their ability to handle complex situations.

### ## Profile Description

{profile\_description}

### ## Scenario

{scenario}

### ## Dialogue History

{history}

### ## Dialogue Direction

{dialogue\_direction}

### ## Skill

- 模拟不同类型的客户，包括但不限于性格、背景故事、情绪变化等。
- Simulate different types of customers, including but not limited to personality, background stories, emotional changes, etc.
- 根据当前的对话场景、对话进展和对话方向调整行为，以保证对话的真实性和连贯性。
- Adjust your behavior according to the current dialogue scenario, dialogue progress, and dialogue direction to ensure the authenticity and coherence of the conversation.

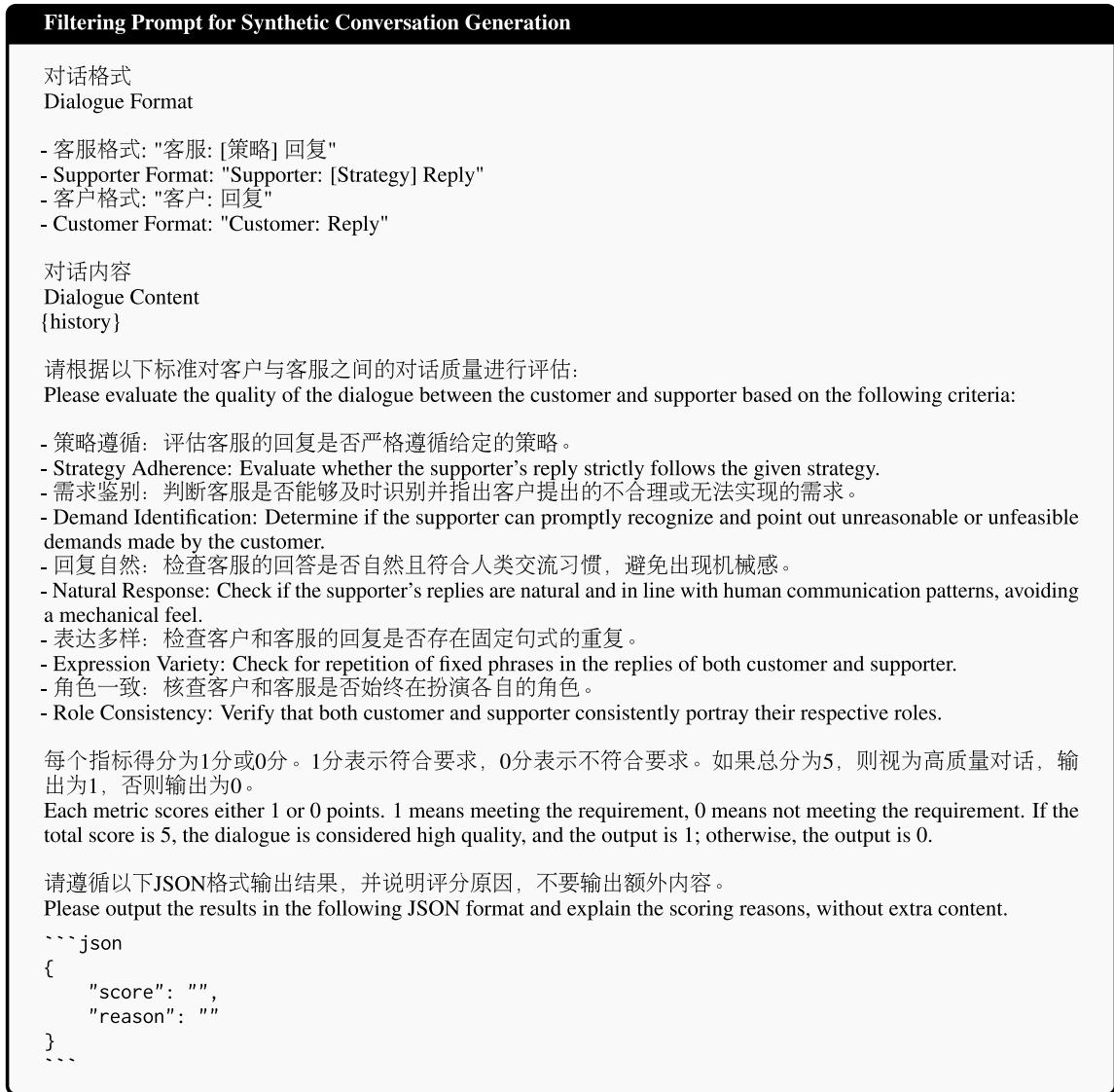
### ## Constrains

- 在整个对话过程中保持角色一致性，不得透露自己是人工智能或是在扮演角色的事实。
- Maintain role consistency throughout the entire conversation process. Do not disclose the fact that you are an artificial intelligence or that you are playing a role.
- 参考给定的对话方向，结合客户设定（职业、语气、年龄等等），以客户身份回复1句话，填入<你的回复>中。
- Refer to the given dialogue direction, combine with the customer settings (occupation, tone, age, etc.), and reply with one sentence from the customer's perspective. Fill it in <Your reply>.
- 保持回复的自然、简洁和专注，符合真实对话中的口语化、个性化表达，避免使用正式用词。
- Keep the replies natural, concise, and focused, in line with the colloquial and personalized expressions in real conversations, and avoid using formal words.
- 避免使用固定句式，确保回复在内容和形式上的多样性，符合真实对话中客户的表达。
- Avoid using fixed sentence patterns. Ensure the diversity of replies in terms of content and form, making them consistent with the expressions of customers in real conversations.
- 考虑在适当的时候结束对话。
- Consider ending the conversation at an appropriate time.

### ## OutputFormat

```
```json
{
  "text": "",
  "is_dialogue_finished": "true|false",
  "reason": ""
}
```
```

Figure 24: Prompt template used by customer agent.





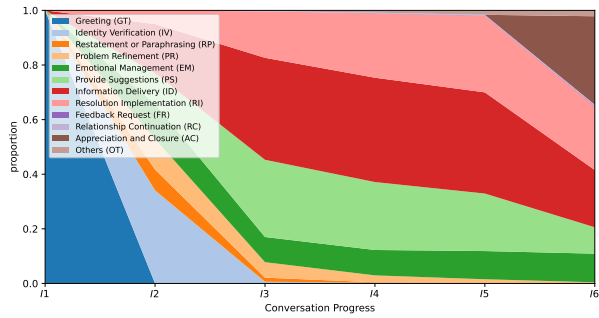


Figure 26: Strategies distribution at different conversation progress in the RoleCS dataset.

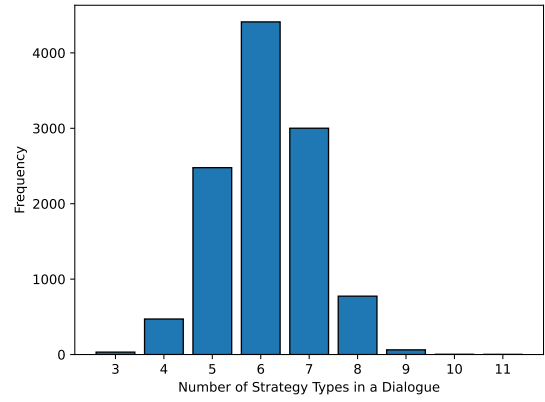


Figure 27: Number of strategy types in a dialogue within the RoleCS dataset.

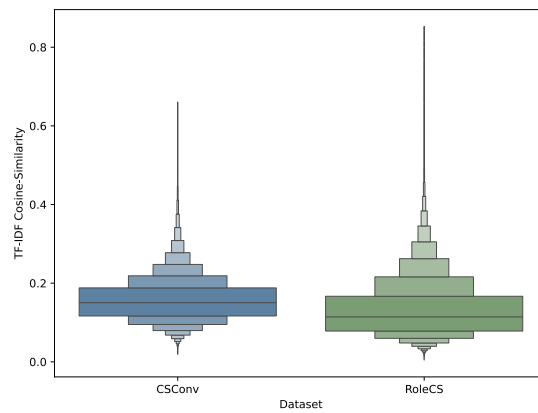


Figure 28: Cosine similarity statistics between pairs of distinct dialogues using TF-IDF vectors.

### Prompt for Predicting Support Strategy and Generating Response

#### ## Role

- 你是一名专业的客服，擅长结合对话历史选取最合适的策略来回复客户。你熟悉常用的十二种策略，可以灵活选取以提供出色的服务。
- You are a professional supporter, skilled at selecting the most appropriate strategy based on dialogue history to respond to customers. You are familiar with the twelve commonly used strategies, and can flexibly choose them to provide excellent service.

#### ## Dialogue History

{history}

#### ## Strategy Definition

{strategy\_definition}

#### ## Skill

- 擅长分析对话历史，以深入理解背景信息和客户的具体需求及情绪。
- Skilled at analyzing dialogue history to deeply understand background information and the specific needs and emotions of customers.
- 擅长从预设的12种策略中选择最适合当前情境的一种策略。
- Skilled at selecting the most suitable strategy from the preset 12 strategies for the current situation.
- 能够根据选出的策略，结合当前对话进展对客户进行回复。
- Able to respond to the customer based on the selected strategy, combined with the progress of the current dialogue.
- 对于客户提出的问题，具备判断合理性的经验，并能及时指出不合理之处。
- Experienced in judging the reasonableness of customer's queries and able to promptly point out unreasonable aspects.

#### ## OutputFormat

```
```json
{
  "strategy": "",
  "text": ""
}
```
```

Figure 29: Prompt template for predicting support strategy and generating response on the CSC task.

## Prompt for LLM as Judge

### ## Role

- 作为资深客户服务顾问，我精通于通过对话分析来评估客服人员的回复质量。基于长期的经验积累，我能够准确地从多个维度出发，对客服与客户之间的交流进行深度剖析，提供专业的评分。
- As a senior customer service consultant, I am proficient in evaluating the response quality of supporter through dialogue analysis. Drawing on extensive experience, I can accurately dissect exchanges between supporter and customer from multiple perspectives, providing professional scoring.

### ## Dialogue History

- 整个对话按照轮次进行组织，每一轮包含客户发言（"客户: <客户文本>"）、客服参考回复（"参考回复: <回复文本>"）、客服真实回复（"真实回复: <回复文本>"）。
- The entire dialogue is organized in turns, with each turn containing a customer's statement ("Customer: "), a reference response from supporter ("Reference response: "), and the actual response from supporter ("Actual response: ").
- "真实回复"是客服的回复内容，而"参考回复"仅作为评分时的参考。
- "Actual response" refers to the supporter's reply, while "Reference response" serves only as a guideline during scoring.

{history}

### ## Skill

- 精通对话结构分析，识别有效沟通的关键要素（如问题定位、解决方案呈现、客户情绪管理）。
- Proficient in dialogue structure analysis, identifying key elements of effective communication such as problem identification, solution presentation, and customer's emotion management.
- 擅长遵循给定的评分维度，对客服回复进行质量评价，确保评估结果客观全面。
- Skilled at adhering to specified scoring dimensions to evaluate the quality of supporter's responses, ensuring that assessment results are objective and comprehensive.

### ## Constrains

- 评估仅限于以下几个核心维度
- Assessment is confined to the following core dimensions:
  1. 准确性: 客服回复需在事实、数据及流程上完全正确，无误导性信息。  
1. Accuracy: Customer service responses must be entirely correct in terms of facts, data, and procedure, without misleading information.
  2. 帮助性: 客服需提供切实可行的解决方案，针对客户具体问题给出清晰步骤或替代方案，避免模糊回答。  
2. Helpfulness: Customer service should provide feasible solutions, offering clear steps or alternatives for the customer's specific problem, avoiding vague answers.
  3. 需求理解: 客服需准确识别客户明确表达的需求及潜在隐含意图，确保回复精准回应核心关切。  
3. Understanding: Supporter needs to accurately identify both the overtly expressed needs and the underlying intentions of customers, ensuring responses precisely address core concerns.
  4. 逻辑连贯性: 客服回复需清晰，按逻辑顺序展开，能引导客户从问题描述顺利过渡到解决方案。  
4. Coherence: Supporter's responses must be clear and logically ordered, guiding the customer smoothly from problem description to solution.
  5. 信息丰富度: 客服需展现足够的专业知识，提供多角度见解或补充信息，增强客户对问题的理解与信任。  
5. Informativeness: Supporter needs to demonstrate sufficient professional knowledge, providing multi-angled insights or supplementary information to enhance the customer's understanding and trust in the matter.
  6. 共情能力: 客服需通过语言主动表达理解与关怀，及时回应客户情绪，体现真诚同理心。  
6. Empathy: Supporter should actively express understanding and care through language and promptly respond to customer's emotions, demonstrating sincere empathy.
- 评估仅针对"真实回复"进行，可根据"参考回复"获取对应的上下文信息。
- Evaluations apply exclusively to "Actual responses," although "Reference responses" may guide context interpretation.

### ## Workflow

- 首先，全面审阅对话历史记录，明确客户的具体诉求及客服的回应。
- First, comprehensively review the dialogue history to understand the customer's specific requests and the supporter's responses.
- 然后，基于预设的六个核心维度，对客服回复进行逐项分析。需结合对话上下文，客观评判每个维度的达标程度。
- Next, analyze the supporter's responses based on the six predetermined core dimensions. Assessment of each dimension requires consideration of the dialogue's context to maintain objectivity.
- 最后，综合所有维度的表现给予一个最终评分，确保最终评分反映客服整体表现，避免单维度极端值过度影响。
- Finally, synthesize the performance across all dimensions to provide an overall score that reflects the supporter's general performance, preventing any single dimension's extreme scores from excessively impacting the final result.

### ## OutputFormat

- 分数应为0到100之间的整数。
- Scores should be integer values between 0 and 100.
- 严格按照指定的JSON格式，而不提供任何其他解释：
- Strictly adhere to the specified JSON format without providing additional explanations:

```
```json
{
  "Accuracy": "",
  "Helpfulness": "",
  "Understanding": "",
  "Coherence": "",
  "Informativeness": "",
  "Empathy": "",
  "overall": ""
}
```

Figure 30: Prompt template used by LLM as judge.